Supporting Information: Supporting Information: Conditioning Protein Generation via Hopfield Pattern Multiplicity

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Sequence Encoding, PCA Projection, and Multiplicity Conditioning

Stochastic attention (SA) operates on continuous vectors on the unit sphere, while protein sequences are discrete objects over a 20-letter amino acid alphabet. Bridging the two requires three transformations—one-hot encoding, PCA dimensionality reduction, and unit-norm projection—and the reverse path (inverse PCA followed by argmax decoding) recovers discrete sequences from continuous samples. Here we describe the full encoding–decoding pipeline, derive the multiplicity-weighted conditioning formulas used in the main text, and discuss how the multiplicity vector $\mathbf{r}$ interacts with the PCA representation.

Given a cleaned alignment of K sequences, each of length L amino acids over the standard 20-letter alphabet $\mathcal{A} = \{\mathrm{A}, \mathrm{R}, \mathrm{N}, \ldots, \mathrm{V}\}$, we encoded each sequence as a binary vector in $\mathbb{R}^{20L}$ by concatenating per-position indicator vectors. For sequence k ($k = 1, \ldots, K$) with

amino acid $a_\ell^{(k)}$ at alignment position ℓ ($\ell = 1, \ldots, L$):

$$\mathbf{x}_k = \left[\mathbf{e}_{a_1^{(k)}},\ \mathbf{e}_{a_2^{(k)}},\ \ldots,\ \mathbf{e}_{a_L^{(k)}}\right]^\top \in \{0,1\}^{20L}, \tag{S1}$$

where $\mathbf{e}_a \in \{0,1\}^{20}$ is the standard basis vector for amino acid a (i.e., a one-hot indicator with a 1 in the position corresponding to a and 0 elsewhere). The full one-hot dimensionality is $d_{\text{full}} = 20L$, which ranged from 520 (ω-conotoxin, L=26) to 1,060 (Kunitz, L=53) across the four families. This space is vastly over-dimensioned relative to the number of stored patterns K (d_{full}/K ranged from 7.0 to 10.7), which creates a problem for the modern Hopfield energy:[?] the similarity score $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ between a stored pattern $\mathbf{m}_k$ and a query state $\boldsymbol{\xi}$ has variance $1/d$ for a random query on the unit sphere $\mathbb{S}^{d-1}$, so when d is large all scores concentrate near zero and the inverse temperature β must be set extremely high to differentiate among patterns, making Langevin sampling inefficient. PCA removes this redundancy by projecting onto the $d \ll d_{\text{full}}$ directions of actual variation in the alignment.

We computed PCA from the one-hot encoded alignment by first centering the data: $\bar{\mathbf{x}} = K^{-1}\sum_{k=1}^{K}\mathbf{x}_k$ (the family's position-specific amino acid frequency profile), $\tilde{\mathbf{X}} = \mathbf{X} - \bar{\mathbf{x}}\mathbf{1}_K^\top$, and then computing the economy SVD $\tilde{\mathbf{X}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, where $\mathbf{U} \in \mathbb{R}^{d_{\text{full}} \times r}$ contains the principal directions of variation, $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \ldots, \sigma_r)$ the singular values in decreasing order, and $r = \text{rank}(\tilde{\mathbf{X}}) \leq \min(d_{\text{full}}, K) - 1$. We retained the smallest dimension d such that $\sum_{j=1}^{d}\sigma_j^2 / \sum_{j=1}^{r}\sigma_j^2 \geq 0.95$ (at least 95% of total variance). Across the four families, d ranged from 20 (ω-conotoxin) to 186 (WW), yielding compression ratios of 3.3× to 26× (Table **??**). Each sequence was then projected as $\mathbf{z}_k = \mathbf{W}_d^\top(\mathbf{x}_k - \bar{\mathbf{x}}) \in \mathbb{R}^d$, where $\mathbf{W}_d = \mathbf{U}_{:,1:d}$ is the projection matrix, and normalized to unit L_2 norm: $\mathbf{m}_k = \mathbf{z}_k / \|\mathbf{z}_k\|_2 \in \mathbb{S}^{d-1}$. This normalization ensures that similarity scores lie in $[-1, 1]$ and prevents patterns with larger PCA norms from dominating the softmax attention. The columns $\mathbf{m}_1, \ldots, \mathbf{m}_K$ form the memory matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d \times K}$ used throughout the paper. The Langevin sampler (Eq. **??**) produces continuous states $\boldsymbol{\xi} \in \mathbb{R}^d$, which are decoded back to amino acid sequences by

inverse PCA reconstruction $\hat{\mathbf{x}} = \bar{\mathbf{x}} + \mathbf{W}_d \boldsymbol{\xi} \in \mathbb{R}^{d_{\text{full}}}$ followed by per-position argmax: $a_\ell = \arg\max_{a \in \mathcal{A}} \ \hat{x}_{20(\ell-1)+a}$ for $\ell = 1, \ldots, L$. This decoding is deterministic and produced valid amino acids at 100% of positions across all families and conditions.

The multiplicity-weighted Hopfield energy (Eq. **??**) assigns a weight $r_k > 0$ to each stored pattern $\mathbf{m}_k$. For the binary functional split used throughout this paper, we set $r_k = \rho$ for the K_{des} designated patterns and $r_k = 1$ for the $K_{\text{bg}} = K - K_{\text{des}}$ background patterns, where $\rho \geq 1$ is the multiplicity ratio. The weighted score function (Proposition **??**) adds $\log \rho$ to the pre-softmax logits of designated patterns and $\log 1 = 0$ to background patterns. To derive the effect of ρ on the attention distribution, consider the idealized case where the query $\boldsymbol{\xi}$ has equal similarity to all stored patterns, i.e., $\mathbf{m}_k^\top \boldsymbol{\xi} = c$ for all k. The softmax attention weight on pattern k simplifies to $a_k = r_k / \sum_j r_j$, and the total attention weight on all designated patterns is:

$$f_{\text{eff}}(\rho) \ = \ \sum_{k \in \text{des}} a_k \ = \ \frac{K_{\text{des}}\, \rho}{K_{\text{des}}\, \rho + K_{\text{bg}}}. \tag{S2}$$

At $\rho = 1$, $f_{\text{eff}} = K_{\text{des}}/K$ (the natural proportion); as $\rho \to \infty$, $f_{\text{eff}} \to 1$. To find the multiplicity ratio needed for a target effective fraction $f \in (0, 1)$, we invert Eq. (S2) by multiplying both sides by the denominator and solving for ρ:

$$\begin{aligned} f\,(K_{\text{des}}\, \rho + K_{\text{bg}}) &= K_{\text{des}}\, \rho, \\ f\, K_{\text{bg}} &= K_{\text{des}}\, \rho\, (1 - f), \\ \rho(f) &= \frac{f\, K_{\text{bg}}}{K_{\text{des}}\, (1 - f)}. \end{aligned} \tag{S3}$$

This is Eq. (**??**) in the main text. The formula is exact under the equal-similarity assumption; in practice, we confirmed empirically that the mean attention weight on designated patterns tracked $f_{\text{eff}}(\rho)$ with $< 0.3\%$ deviation across all ρ values and all families (Table **??**), validating the approximation as operationally exact. The effective number of patterns under

multiplicity weighting, $K_{\text{eff}}(\mathbf{r}) = (\sum_k r_k)^2 / \sum_k r_k^2 = (K_{\text{des}}\,\rho + K_{\text{bg}})^2/(K_{\text{des}}\,\rho^2 + K_{\text{bg}})$, equals K when $\rho = 1$ and converges to K_{des} as $\rho \to \infty$; this quantity governs the shift in the phase transition temperature β^*.

A key property of multiplicity-weighted conditioning is that $\mathbf{r}$ does *not* alter the PCA encoding. The memory matrix $\hat{\mathbf{X}}$ is constructed once from the full family alignment and remains fixed regardless of ρ; the multiplicity weights enter only through the softmax logits. This separation has three consequences. First, the PCA subspace is independent of ρ: the principal directions $\mathbf{W}_d$, the centering vector $\bar{\mathbf{x}}$, and the retained dimension d are all determined by the alignment alone, so the calibration gap $\Delta = f_{\text{eff}} - f_{\text{obs}}$ is identical for all ρ values on a given family. Second, ρ controls attention, not geometry: increasing ρ adds $\log \rho$ to the designated logits, shifting attention weight toward designated patterns, but this shift must still pass through the PCA reconstruction and argmax decoding to produce discrete residues, which is where the calibration gap arises. Third, ρ shifts the phase transition: the logit biases pre-concentrate the attention distribution at $\beta = 0$, reducing the weighted entropy to $H_{\mathbf{r}}(0) = \log K_{\text{eff}}(\mathbf{r}) < \log K$, so reaching the retrieval-phase inflection requires a higher inverse temperature $\beta^*(\rho) \geq \beta^*(1)$. We observed this empirically: β^* increased from 4.3 ($\rho = 1$, $K_{\text{eff}} = 99$) to 10.1 ($\rho = 1{,}000$, $K_{\text{eff}} = 32.1$) on the Kunitz domain (Fig. **??**). By contrast, hard curation—where the memory matrix is built from the designated subset only—*does* change the PCA: the centering vector, principal directions, and retained dimension are all computed from K_{des} sequences, capturing subset-specific variation that may be orthogonal to the leading components of the full-family PCA. This is why hard curation achieved 100% phenotype transfer regardless of the Fisher separation index S, while multiplicity weighting was limited by the full-family PCA geometry.

Supplementary Tables and Figures

Tables S1–S3 report selected values from the ρ sweep on each of the three benchmark families. f_{eff} is the effective designated fraction computed from the multiplicity vector; f_{obs} is the hard-decoded marker fraction averaged over 930 generated sequences; $\bar{a}_{\text{des}}$ is the mean softmax attention weight on designated patterns; and D is the mean pairwise diversity. Table S4 reports phenotype fraction and diversity as a function of β/β^* for three fixed ρ values on the Kunitz domain. Figure S1 shows the joint distribution of ESMFold pLDDT and TM-score for all 250 predicted Kunitz structures: SA-generated sequences cluster tightly in the high-pLDDT, high-TM-score region while HMM-emitted sequences scatter broadly across lower values. The entropy curves for the full ρ sweep ($\rho \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 1000\}$) on the Kunitz family produce a monotonically rightward-shifting family of sigmoid curves with $\beta^*(\rho)$ increasing from 3.85 ($\rho = 1$) to 10.22 ($\rho = 1000$), consistent with $\beta^* \propto \log K_{\text{eff}}$ predicted by the mean-field argument in Section **??** (Fig. **??**).

Table S1: Kunitz domain (PF00014) ρ sweep. $K = 99$, $K_{\text{des}} = 32$.

ρ	f_{eff}	f_{obs}	$\bar{a}_{\text{des}}$	D
1	0.323	0.406	0.324	0.563
5	0.705	0.481	0.712	0.556
10	0.827	0.539	0.832	0.547
50	0.962	0.598	0.963	0.526
100	0.980	0.615	0.981	0.520
200	0.990	0.623	0.991	0.519
500	0.996	0.631	0.997	0.515
1000	0.998	0.634	0.998	0.514

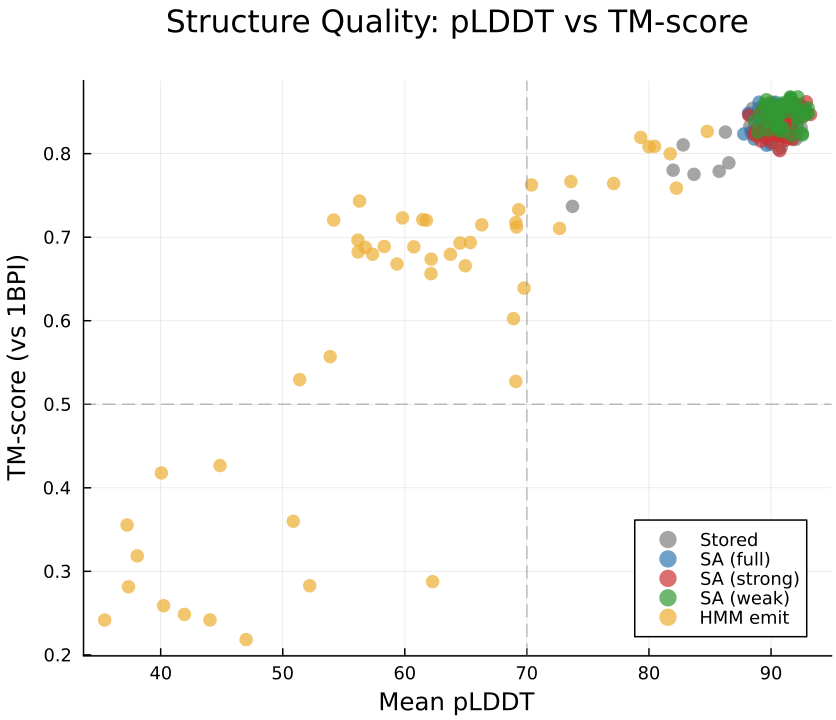

Figure S1: **pLDDT and TM-score are positively correlated; SA sequences cluster in the high-quality region.** Each point represents one predicted structure (50 per source). Stored (gray) and SA-generated (blue, red, green) sequences cluster tightly above pLDDT > 85 and TM-score > 0.8 (upper right quadrant). HMM sequences (orange) scatter broadly, with many falling below the pLDDT > 70 confidence threshold (vertical dashed line) and the TM-score > 0.5 same-fold threshold (horizontal dashed line).

Table S2: SH3 domain (PF00018) ρ sweep. $K = 55$, $K_{\text{des}} = 33$.

ρ	f_{eff}	f_{obs}	$\bar{a}_{\text{des}}$	D
1	0.600	0.852	0.614	0.580
5	0.882	0.897	0.890	0.575
10	0.938	0.906	0.941	0.575
50	0.983	0.965	0.983	0.567
100	0.991	0.977	0.991	0.559
200	0.996	0.984	0.996	0.554
500	0.999	0.989	0.999	0.513
1000	0.999	0.992	0.999	0.510

Table S3: WW domain (PF00397) ρ sweep. $K = 420$, $K_{\text{des}} = 69$.

ρ	f_{eff}	f_{obs}	$\bar{a}_{\text{des}}$	D
1	0.164	0.131	0.152	0.690
5	0.496	0.184	0.487	0.689
10	0.663	0.226	0.661	0.676
50	0.901	0.277	0.901	0.657
100	0.947	0.303	0.947	0.649
200	0.973	0.327	0.974	0.643
500	0.990	0.360	0.991	0.642
1000	0.995	0.371	0.995	0.642

Table S4: β sweep on Kunitz domain ($K = 99$, $K_{\text{des}} = 32$).

ρ	β/β^*	f_{obs}	D
10	0.50	0.471	0.581
10	1.00	0.539	0.547
10	1.50	0.573	0.525
10	2.00	0.597	0.508
10	3.00	0.613	0.484
50	0.50	0.506	0.573
50	1.00	0.582	0.537
50	1.50	0.631	0.514
50	2.00	0.653	0.499
50	3.00	0.700	0.469
200	0.50	0.535	0.564
200	1.00	0.613	0.527
200	1.50	0.648	0.503
200	2.00	0.677	0.488
200	3.00	0.706	0.463